Tumor NeoAntigen Profiling with Validated Patient-Specific NeoTCR Characterization to Improve NeoEpitope Prediction

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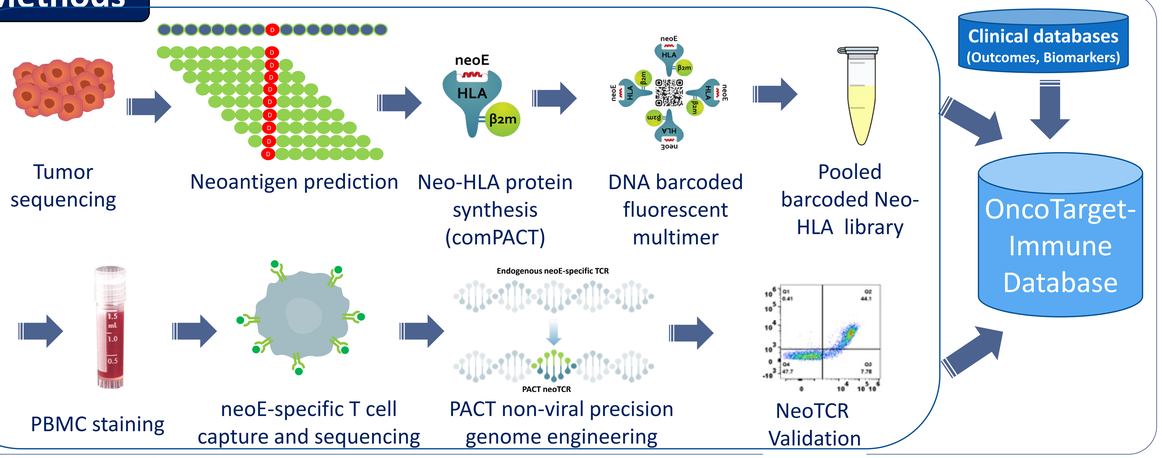
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Abstract

PACT Pharma has developed an ultra-sensitive approach to validate predicted tumor-exclusive neoantigens, HLAs and the cognate T cell receptors (neoTCRs) by capturing neoantigen-specific T cells from peripheral blood. This neoTCR discovery and validation process is being applied in a clinical trial (NCT03970382) evaluating personalized neoTCR-T cell therapy to treat persons across eight different solid tumor types. Using our state-of-the-art pipeline and screening process, more than 100 neoantigens were predicted from patients across >5 types of solid cancers to date and concomitantly validated by capturing CD8 T cells from patient blood. We discovered and characterized ~200 unique neoTCRs that specifically bound to the predicted neoE-HLA tumor targets. These validated neoepitopes (neoE) of 8 to 11 amino acids represent broad HLA class I coverage with >30 alleles to date. Our analysis revealed that the single amino acid mutations occur in all positions within the validated neoE peptide. Epitope immunogenicity resulting in immune recognition is affected by interactions between mutated residues to the HLA receptor and to their cognate TCRs as well as by neoantigen expression levels. Our results revealed that the validated neoepitopes appear to exhibit broad ranges of predicted HLA binding affinities, neoTCR binding stability and neoantigen expression levels. The analysis presented here offers insights to enable machine learning to advance rules for epitope selection and prioritization that may be important for immunological approaches to address a broad range of diseases, including for patients with solid cancers.

Methods



Conclusions

- **Neoepitope-HLA target predictions resulting in validated neoTCRs were interpreted to represent verified tumor-displayed targets to which that patient mounted an immune response;**
- **The PACT neoTCR discovery and validation platform produces high confidence predictions, resulting in 3.2% of patient neoE-HLA targets with a confirmed neoTCR in 32 HLA alleles, demonstrating robust and broad applicability of the technology; immunogenicity and tumor target validation does not appear to be strongly correlated with predicted binding affinity;**
- **Through this unbiased screening work, the results demonstrate that mutated residues can be at any position within neoepitope peptides (neoE);**
- **Trends in binding affinity & stability consistent with previous reports were observed amongst neoEs resulting in neoTCRs. However, peptide foreignness differences were not observed in this dataset, in contrast to previous reports[1];**
- **The PACT OncoTarget-Immune Database of neoE predictions, neoTCR validations and clinical data represents a significant & growing dataset for patient-specific tumor immunogenicity in solid cancers; Future improvements in heuristics for epitope selection & prioritization are underway;**

References: [1] *Wells et al., 2020, Cell 183, 1–17*

Results

PACT's neoTCR discovery pipeline successfully identified & validated >200 neoTCRs targeting cancer-specific neoepitopes presented by >30 different HLA alleles across >5 different solid tumor types

	With validated neoTCR
Patients	43
HLA Alleles	32
Unique neoE-HLA with TCRs	118
TCRs	211

Table 1. Summary from current OncoTarget-Immune Database analysis.

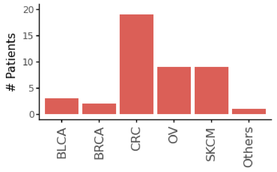


Figure 1. Distribution of PACT Patients by cancer type.

PACT's predictions of high confidence screening targets resulted in 3.2% (118/3714) of neoE-HLAs with immunogenic and validated neoTCRs

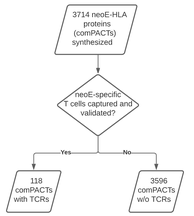


Figure 2. Overview of analysis groups. 3714 neoE-HLA proteins were successfully synthesized. These were categorized as having neoE-specific T cells captured and validated (118) or not (3596).

The majority of neoE peptide targets with validated neoTCRs consisted of 9-mer or 10-mer peptides with single amino acid mutations found at each position

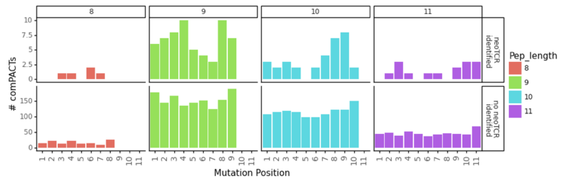


Figure 3. NeoE-HLA grouped by peptide length and mutation position. Peptides from 8-11mers were predicted through the pipeline that resulted in a cognate neoTCR (top) or no neoTCR (bottom).

An overall trend in predicted stability was observed for cognate TCR neoE

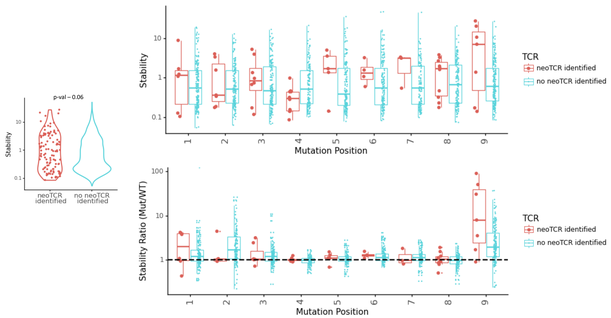


Figure 6. Composite NeoE-HLA predicted stabilities (left), predicted stabilities for mutated positions for 9mers (top right) and stability ratios (Mut/WT) for mutated positions for 9 mers (bottom right). NeoE-HLA stabilities were predicted by netMHCstabpan for the half-life of the complexes. The stability ratio was calculated by dividing neoE-HLA stability by the stability of its wild-type epitope and HLA complex.

Only mutations at position 9 resulted in a significant increase in predicted peptide-HLA binding affinity

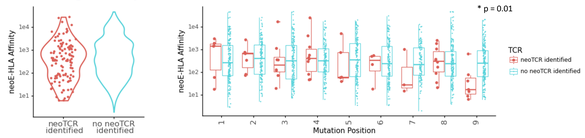


Figure 4. Composite NeoE-HLA predicted binding affinities (left) and predicted binding affinities for mutated positions for 9mers (right). Binding affinities were predicted by netMHCpan.

A trend towards lower agretopicity was observed at positions 1,2 and 9 of 9-mer peptides

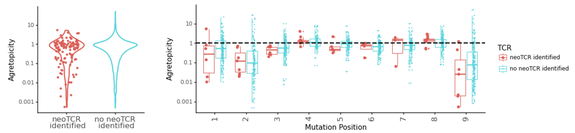


Figure 5. (left panel) Composite predicted agretopicity (namely, mutated binding affinity/wild-type binding affinity) for immunogenic peptides (with validated neoTCR) and for other predicted peptide-HLA targets (without validated neoTCR); and (right panel) predicted agretopicity for all mutated positions for 9mers.

No significant differences in peptide "foreignness" were observed

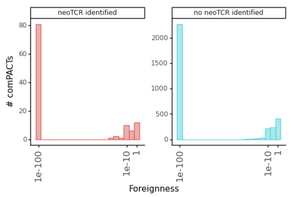


Figure 7. Distribution of predicted foreignness for neoE-HLA target for which a neoTCR was validated (left) and all other predicted neoE-HLA targets (without a validated TCR, right panel).